Remember

When multiplying or dividing integers:

1. Like signs give a plus
2. Unlike signs give a minus

Example 2

Write as a single integer:

(i) -4×6 (ii) -7×-2 (iii) $-5 \times -8 \times 2$

(i) $-4 \times 6 = -24$ (*unlike signs give minus*)
(ii) $-7 \times -2 = +14$ (*like signs give plus*)
(iii) $-5 \times -8 \times 2 = 40 \times 2 = 80$ (*take two terms at a time*)

Example 3

Write as a single integer:

(i) $\frac{18}{3}$ (ii) $\frac{-20}{-5}$ (iii) $\frac{18}{-9}$ (iv) $\frac{-15}{5}$

(i) $\frac{18}{3} = 6$ (ii) $\frac{-20}{-5} = 4$ (*like signs give plus*)
(iii) $\frac{18}{-9} = -2$ (iv) $\frac{-15}{5} = -3$ (*unlike signs give minus*)

Exercise 1B

1. What temperature is

(i) 3°C Warmer than 7°C
(ii) 5°C colder than 8°C
(iii) 6°C colder than 3°C
(iv) 3°C colder than −4°C
(v) 4°C warmer than −10°C
(vi) 8°C warmer than −8°C?

2. Say if each of the following is true of false:

(i) $9 > 0$ (ii) $-1 < 4$ (iii) $2 < -3$ (iv) $-4 > -1$
(v) $0 > -4$ (vi) $-12 > 6$ (vii) $-1 < -7$ (viii) $6 > -7$

3. Insert the symbol $>$ or $<$ between each of the following pairs of numbers so as to make the statements true:

(i) 6 7 (ii) 4 −1 (iii) 0 −5
(iv) −5 4 (v) −2 −5 (vi) −10 −7
(vii) −8 0 (viii) −3 −20 (ix) −2 1

4. Express each of the following as a single integer:

(i) $6 + 7 - 5$ (ii) $12 + 4 - 8$ (iii) $14 - 3 - 6$
(iv) $3 - 7 + 9$ (v) $6 - 2 - 9$ (vi) $-3 + 7 + 5$
(vii) $-3 - 4 + 14$ (viii) $-2 - 7 + 16$ (ix) $3 - 12 + 15$

5. Express as single integers:

(i) $12 - 9 + 14 - 7$ (ii) $12 - 8 + 9 - 4$

(iii) $7 - 3 - 9 + 12$ (iv) $14 - 8 + 6 - 7 + 2$

6. Find the missing integer in each of the following boxes:

(i) $\square + 6 = 9$ (ii) $7 + \square = 2$ (iii) $\square - 6 = -8$

(iv) $9 + \square = -3$ (v) $-9 - (-8) = \square$ (vi) $-6 - \square = 14$

7. Write down the answer to each of the following:

(i) 6×4 (ii) -5×4 (iii) $7 \times (-8)$

(iv) $5 \times (-9)$ (v) -6×9 (vi) $(-9) \times (-4)$

(vii) $(-8) \times (-6)$ (viii) $10 \times (-7)$ (ix) -9×8

(x) $(-6) \times (-11)$ (xi) $4 \times (-20)$ (xii) $(-18) \times 3$

8. Express as a single integer:

(i) $6 \times 2 \times (-3)$ (ii) $4 \times (-3) \times (-2)$ (iii) $(-6) \times 5 \times (-3)$

(iv) $(-4) \times (-3) \times 5$ (v) $(-2) \times 6 \times (-8)$ (vi) $(-3) \times (-2) \times (-7)$

9. Divide each of the following:

(i) $\frac{16}{2}$ (ii) $\frac{-8}{4}$ (iii) $\frac{-27}{3}$ (iv) $\frac{15}{-3}$ (v) $\frac{-42}{-7}$

(vi) $\frac{-35}{7}$ (vii) $\frac{54}{-6}$ (viii) $\frac{-54}{-9}$ (ix) $\frac{-63}{7}$ (x) $\frac{55}{-11}$

10. Express each of the following as a single integer:

(i) $\frac{3 \times (-6)}{9}$ (ii) $\frac{(-5) \times (-10)}{-10}$ (iiii) $\frac{(-12) \times 4}{8}$

(iv) $\frac{(-8) \times (-9)}{-12}$ (v) $\frac{9 \times (-6)}{-2}$ (vi) $\frac{(-8) \times 3 \times (-6)}{-12}$

11. Noting that you multiply and divide before you add and subtract, simplify each of the following:

(i) $3 - 8 + 8 \times (-2)$ (ii) $15 + 6 \div 2 + 12 \times (-2)$

(iii) $(16 - 4) \div 2 + 5 \times (-4)$ (iv) $(15 - 10) \times (-3) + 5 \times (-6)$

(v) $14 + 4 \times (-2) - (7 - 15)$ (vi) $100 \div (13 - 3) + (-4) \times (-7)$

12. Find the missing integer in each of the following boxes:

(i) $\frac{\square}{4} = 4$ (ii) $\frac{5}{\square} = -1$ (iii) $\frac{\square}{3} = -4$ (iv) $\frac{\square}{-3} = -4$

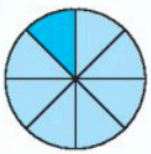
Section 1C Fractions (Rational Numbers)

Simplifying Fractions

When we think of fractions we think of parts of a whole.

Take the fraction $\frac{2}{6}$, this gives two parts from six equal parts.

(a)

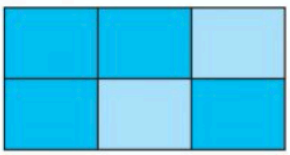


$\left(\frac{1}{8}\right)$

(b)

$\left(\frac{2}{6}\right)$

(c)



$\left(\frac{4}{6}\right)$